

OLMESAR 20 (Olmesartan Medoxomil Tablets 20mg)
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Name of the product

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Name and strength of Active Ingredient

Name : Olmesartan Medoxomil

Strength : 20mg & 40 mg

Product Description

Olmesartan Medoxomil Tablets 20mg

White to off white round, biconvex film coated tablets debossed with 'H' on one side and 'O3' on the other side.

Olmesartan Medoxomil Tablets 40mg

White to off white oval, biconvex film coated tablets debossed with 'H' on one side and 'O4' on the other side.

Dosage form:

Tablet in the form of film coated

Pharmacodynamics/Pharmacokinetics

Pharmacodynamics:

Pharmacology:

Mechanism of Action: Olmesartan medoxomil is a prodrug, which is hydrolysed to the active metabolite olmesartan during absorption from the gastrointestinal tract. Olmesartan is a selective AT₁ subtype angiotensin II receptor antagonist.

Angiotensin II is formed from angiotensin I in a reaction catalyzed by angiotensin-converting enzyme (ACE, kininase II). Angiotensin II is the principal pressor agent of the renin-angiotensin system, with effects that include vasoconstriction, stimulation of synthesis and release of aldosterone, cardiac stimulation and renal reabsorption of sodium. Olmesartan blocks the vasoconstrictor effects of angiotensin II by selectively blocking the binding of angiotensin II to the AT₁ receptor in vascular smooth muscle. Its action is, therefore, independent of the pathways for angiotensin II synthesis.

An AT₂ receptor is found also in many tissues, but this receptor is not known to be associated with cardiovascular homeostasis. Olmesartan has more than a 12,500-fold greater affinity for the AT₁ receptor than for the AT₂

receptor. Blockade of the renin-angiotensin system with ACE inhibitors, which inhibit the biosynthesis of angiotensin II from angiotensin I, is a mechanism of many drugs used to treat hypertension. ACE inhibitors also inhibit the degradation of bradykinin, a reaction also catalyzed by ACE. Because olmesartan medoxomil does not inhibit ACE (kininase II), it does not affect the response to bradykinin.

Whether this difference has clinical relevance is not yet known. Blockade of the angiotensin II receptor inhibits the negative regulatory feedback of angiotensin II on renin secretion, but the resulting increased plasma renin activity and circulating angiotensin II levels do not overcome the effect of olmesartan on blood pressure.

Pharmacodynamics: Olmesartan medoxomil doses of 2.5-40 mg inhibit the pressor effects of angiotensin I infusion. The duration of the inhibitory effect was related to dose, with doses of olmesartan medoxomil >40 mg giving >90% inhibition at 24 hrs.

Olmesartan medoxomil once daily lowered diastolic and systolic blood pressure. Olmesartan medoxomil doses >40 mg had little additional effect. The onset of antihypertensive effect occurred within 1 week and was largely manifested after 2 weeks.

The blood pressure-lowering effect was maintained throughout the 24-hr period with olmesartan medoxomil once daily, with trough-to-peak ratios for systolic and diastolic response between 60% and 80%.

Pharmacokinetics

General: Olmesartan medoxomil is rapidly and completely bioactivated by ester hydrolysis to olmesartan during absorption from the gastrointestinal tract. Olmesartan appears to be eliminated in a biphasic manner with a terminal elimination half-life of approximately 13 hrs.

Olmesartan shows linear pharmacokinetics following single oral doses of up to 320 mg and multiple oral doses of up to 80 mg. Steady-state levels of olmesartan are achieved within 3-5 days and no accumulation in plasma occurs with once-daily dosing.

The absolute bioavailability of olmesartan is approximately 26%. After oral administration, the peak plasma concentration (C_{max}) of olmesartan is reached after 1-2 hrs. Food does not affect the bioavailability of olmesartan.

Metabolism and Excretion: Following the rapid and complete conversion of olmesartan medoxomil to olmesartan during absorption, there is virtually no further metabolism of olmesartan. Total plasma clearance of olmesartan is 1.3 L/hr, with a renal clearance of 0.6 L/hr. Approximately 35-50% of the absorbed dose is recovered in urine while the remainder is eliminated in feces via the bile.

Distribution: The volume of distribution of olmesartan is approximately 17 L. Olmesartan is highly bound to plasma proteins (99%) and does not penetrate red blood cells. The protein-binding is constant at plasma

olmesartan concentrations well above the range achieved with recommended doses.

In rats, olmesartan crossed the blood-brain barrier poorly, if at all. Olmesartan passed across the placental barrier in rats and was distributed to the fetus. Olmesartan was distributed to milk at low levels in rats.

Special Populations: *Paediatric population:* The pharmacokinetics of olmesartan have not been investigated in patients <18 years.

Elderly: The pharmacokinetics of olmesartan were studied in the elderly (≥ 65 years). Overall, maximum plasma concentrations of olmesartan were similar in young adults and the elderly. Modest accumulation of olmesartan was observed in the elderly with repeated dosing; $AUC_{ss,\tau}$ was 33% higher in elderly patients, corresponding to an approximate 30% reduction in Cl_R .

Gender: Minor differences were observed in the pharmacokinetics of olmesartan in women compared to men. AUC and C_{max} were 10-15% higher in women than in men.

Renal Insufficiency: In patients with renal insufficiency, serum concentrations of olmesartan were elevated compared to subjects with normal renal function. After repeated dosing, the AUC was approximately tripled in patients with severe renal impairment (creatinine clearance <20 mL/min).

Indications

Olmesar is indicated for the treatment of essential hypertension.

Recommended dose

Dosage must be individualized. The usual recommended starting dose of olmesartan medoxomil is 20 mg once daily when used as monotherapy in patients who are not volume-contracted. If additional blood pressure reduction is required, olmesartan medoxomil dose may be increased to a maximum of 40 mg daily or hydrochlorothiazide therapy may be added.

The antihypertensive effect of olmesartan medoxomil is substantially present within 2 weeks of initiating therapy and is maximal by about 8 weeks after initiating therapy. This should be borne in mind when considering changing the dose regimen for any patient.

Elderly (65 years or over)

No adjustment of dosage is generally required in the elderly (see below for dose recommendations in patients with renal impairment). If up-titration to the maximum dose of 40 mg daily is required, blood pressure should be closely monitored.

Renal impairment

Dosage of olmesartan medoxomil should be individualized in patients with renal impairment.

The maximum dose in patients with mild to moderate renal impairment (creatinine clearance of 20 – 60 mL/min) is 20 mg olmesartan medoxomil once daily, owing to limited experience of higher dosages in this patient group.

The use of olmesartan medoxomil in patients with severe renal impairment (creatinine clearance <20 mL/min) is not recommended, since there is only limited experience in this patient group. There is no experience in the use of olmesartan medoxomil in patients requiring dialysis.

Hepatic impairment

No adjustment of dosage recommendations is required for patients with mild hepatic impairment. In patients with moderate hepatic impairment, an initial dose of 10 mg olmesartan medoxomil once daily is recommended and the maximum dose should not exceed 20 mg once daily. Close monitoring of blood pressure and renal function is advised in hepatically-impaired patients who are already receiving diuretics and/or other antihypertensive agents. There is no experience of olmesartan medoxomil in patients with severe hepatic impairment, therefore use is not recommended in this patient group. Olmesartan medoxomil should not be used in patients with biliary obstruction.

Paediatric population

The safety and efficacy of olmesartan medoxomil in children and adolescents below 18 years has not been established. No data are available. Should not be used in children below 1 years of age because of safety concern and lack of data in this age group.

Method of administration:

In order to assist compliance, it is recommended that olmesartan medoxomil tablets be taken at about the same time each day, with or without food, for example at breakfast time. The tablet should be swallowed with a sufficient amount of fluid (e.g. one glass of water). The tablet should not be chewed.

Route of administration

Oral

Contraindications

Patients who are hypersensitive to any component of Olmesar.

Warnings and precautions

Warnings:

Fetal/Neonatal Morbidity and Mortality: Drugs that act directly on the renin-angiotensin system can cause fetal and neonatal morbidity and death when administered to pregnant women. Several dozen cases have been reported in the world literature of patients who were taking ACE inhibitors. When pregnancy is detected, olmesartan medoxomil should be discontinued as soon as possible.

The use of drugs that act directly on the renin-angiotensin system during the 2nd and 3rd trimesters of pregnancy has been associated with fetal and neonatal injury, including hypotension, neonatal skull hypoplasia, anuria, reversible or irreversible renal failure and death. Oligohydramnios has also been reported, presumably resulting from decreased fetal function; oligohydramnios in this setting has been associated with fetal limb contractures, craniofacial deformation and hypoplastic lung development. Prematurity, intrauterine growth retardation and patent ductus arteriosus have also been reported, although it is not clear whether these occurrences were due to exposure to the drug.

These adverse effects do not appear to have resulted from intrauterine drug exposure that has been limited to the 1st trimester. Mothers whose embryos and fetuses are exposed to an angiotensin II receptor antagonist only during the 1st trimester should be so informed. Nonetheless, when patients become pregnant, physicians should have the patient discontinue the use of olmesartan medoxomil as soon as possible.

Rarely (probably less often than once in every thousand pregnancies), no alternative to a drug acting on the renin-angiotensin system will be found. In these rare cases, the mothers should be apprised of the potential hazards to their fetuses and serial ultrasound examinations should be performed to assess the intra-amniotic environment.

If oligohydramnios is observed, olmesartan medoxomil should be discontinued unless it is considered life-saving for the mother. Contraction stress testing (CST), a nonstress test (NST) or biophysical profiling (BPP) may be appropriate, depending upon the week of pregnancy.

Patients and physicians should be aware, however, that oligohydramnios may not appear until after the fetus has sustained irreversible injury.

Infants with histories of *in utero* exposure to an angiotensin II receptor antagonist should be closely observed for hypotension, oliguria and hyperkalemia. If oliguria occurs, attention should be directed toward support of blood pressure and renal perfusion. Exchange transfusion or dialysis may be required as means of reversing hypotension and/or substituting for disordered renal function.

Hypotension in Volume- or Salt-Depleted Patients: In patients with an activated renin-angiotensin system eg,

volume- and/or salt-depleted patients (eg, those being treated with high doses of diuretics), symptomatic hypotension may occur after initiation of treatment with olmesartan medoxomil. Treatment should start under close medical supervision. If hypotension does occur, the patient should be placed in the supine position and, if necessary, given an IV infusion of normal saline. A transient hypotensive response is not a contraindication to further treatment, which usually can be continued without difficulty once the blood pressure has stabilized.

Driving and using machines

The patient may feel sleepy or dizzy while being treated for high blood pressure. If the tablets make them feel sick, dizzy or tired, or give them a headache, do not drive or use machines.

Special Precautions

General: Impaired Renal Function: As a consequence of inhibiting the renin-angiotensin-aldosterone system, changes in renal function may be anticipated in susceptible individuals treated with olmesartan medoxomil. In patients whose renal function may depend upon the activity of the renin-angiotensin-aldosterone system (eg, patients with severe congestive heart failure), treatment with ACE inhibitors and angiotensin receptor antagonists has been associated with oliguria and/or progressive azotemia and (rarely) with acute renal failure and/or death. Similar results may be anticipated in patients treated with olmesartan medoxomil. **Use in children:** Safety and effectiveness in pediatric patients have not been established. **Use in the elderly:** No overall differences in effectiveness or safety were observed between elderly and younger patients.

Interaction with other Medicaments

The bioavailability of olmesartan was not significantly altered by the co-administration of antacids $[\text{Al}(\text{OH})_3/\text{Mg}(\text{OH})_2]$.

Olmesartan medoxomil is not metabolized by the cytochrome P-450 system and has no effects on P-450 enzymes; thus, interactions with drugs that inhibit, induce or are metabolized by those enzymes are not expected.

Pregnancy and lactation

Category D: There is positive evidence of human foetal risk, but the benefits from use in pregnant women may be acceptable despite the risk (e.g., if the drug is needed in a life-threatening situation or for a serious disease for which safer drugs cannot be used or are ineffective).

Use in pregnancy: Pregnancy Category C (1st trimester) and D (2nd and 3rd trimesters): When used in pregnancy during the 2nd and 3rd trimesters, drugs that act directly on the renin-angiotensin system can cause injury and even death to the developing fetus. When pregnancy is detected, olmesartan medoxomil should be discontinued as soon as possible.

Female patients of childbearing age should be told about the consequences of 2nd and 3rd trimester exposure to drugs that act on the renin-angiotensin system and they should be told also that these consequences do not appear to have resulted from intrauterine drug exposure that has been limited to the 1st trimester. These patients should be asked to report pregnancies to their physicians as soon as possible.

Use in lactation: It is not known whether olmesartan is excreted in human milk. Because of the potential for adverse effects on the nursing infant, a decision should be made whether to discontinue nursing or discontinue the drug, taking into account the importance of the drug to the mother.

Side effects

Summary of the safety profile:

The most commonly reported adverse reactions during treatment are headache, influenza-like symptoms and dizziness.

The following terminologies have been used in order to classify the occurrence of adverse reactions: very common; common; uncommon; rare; very rare.

MedDRA System Organ Class	Adverse reactions	Frequency
Blood and lymphatic system disorders	Thrombocytopenia	Uncommon
Immune system disorders	Anaphylactic reaction	Uncommon
Metabolism and nutrition disorders	Hypertriglyceridaemia	Common
	Hyperuricaemia	Common
	Hyperkalaemia	Rare
Nervous system disorders	Dizziness	Common
	Headache	Common
Ear and labyrinth disorders	Vertigo	Uncommon
Cardiac disorders	Angina pectoris	Uncommon
Vascular disorders	Hypotension	Rare
Respiratory, thoracic and mediastinal disorders	Bronchitis	Common
	Pharyngitis	Common
	Cough	Common
	Rhinitis	Common
Gastrointestinal disorders	Gastroenteritis	Common
	Diarrhoea	Common
	Abdominal pain	Common

	Nausea	Common
	Dyspepsia	Common
	Vomiting	Uncommon
	Sprue-like enteropathy	Very rare
Skin and subcutaneous tissue disorders	Exanthema	Uncommon
	Allergic dermatitis	Uncommon
	Urticaria	Uncommon
	Rash	Uncommon
	Pruritus	Uncommon
	Angioedema	Rare
Musculoskeletal and connective tissue disorders	Arthritis	Common
	Back pain	Common
	Skeletal pain	Common
	Myalgia	Uncommon
	Muscle spasm	Rare
Renal and urinary disorders	Haematuria	Common
	Urinary tract infection	Common
	Acute renal failure	Rare
	Renal insufficiency	Rare
General disorders and administration site conditions	Pain	Common
	Chest pain	Common
	Peripheral oedema	Common
	Influenza-like symptoms	Common
	Fatigue	Common
	Face oedema	Uncommon
	Asthenia	Uncommon
	Malaise	Uncommon
	Lethargy	Rare
Investigations	Hepatic enzymes increased	Common
	Blood urea increased	Common
	Blood creatine phosphokinase increased	Common
	Blood creatinine increased	Rare
Hepatobiliary disorders	Autoimmune hepatitis*	Not known

Single cases of rhabdomyolysis have been reported in temporal association with the intake of angiotensin II receptor blockers.

*Cases of autoimmune hepatitis with a latency of few months to years have been reported post-marketing, that were reversible after the withdrawal of olmesartan.

Additional information on special populations *Paediatric population*

Whilst the nature and severity of the adverse events are similar to that of the adults, the frequency of the following is higher in the children:

- Epistaxis is a common adverse event in children that has not been reported in adults.
 - Dizziness and headache were reported in the high olmesartan medoxomil dose group in children age 6-17 years.
- The overall safety profile for olmesartan medoxomil in paediatric patients does not differ significantly from the safety profile in adults.

Elderly (age 65 years or over)

In elderly people the frequency of hypotension is slightly increased from rare to uncommon.

Symptoms and treatment of overdose

Limited data are available related to overdosage in humans. The most likely manifestations of overdosage would be hypotension and tachycardia; bradycardia could be encountered if parasympathetic (vagal) stimulation occurs. If symptomatic hypotension should occur, supportive treatment should be initiated. The dialyzability of olmesartan is unknown.

Storage condition

Store below 30°C and protect from moisture.

Dosage forms and packaging available

10 x 10's Alu-Alu blister pack.

Product Registration Holder

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